Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A star initially has 10^{40} deuterons. It produces energy via the processes

$$_1^2\mathrm{H} + {}_1^2\mathrm{H} \longrightarrow {}_1^3\mathrm{H} + p$$

and $$_1^2\mathrm{H} + {}_1^3\mathrm{H} \longrightarrow {}_2^4\mathrm{He} + n$$

where the masses of the nuclei are: $m\left({}^2\mathrm{H}\right) = 2.014\mathrm{u}\ m(p) = 1.007\mathrm{u}, m(n) = 1.008\mathrm{u}$ and $m\left({}^4\mathrm{He}\right) = 4.001\mathrm{u}$

If the average power radiated by the star is $10^{16}\ \mathrm{W}$, the deuteron supply of the star is exhausted in a time of the order of

(A) $10^6\ \mathrm{s}$

(B) $10^8\ \mathrm{s}$

(C) $10^{12}\ \mathrm{s}$

(D) $10^{16}\ \mathrm{s}$

Q.2

Binding energy per nucleon versus mass number curve for nuclei is shown in Fig. 19.2. W, X, Y and Z are four nuclei indicated on the curve. The process that would release energy is

(A) $Y \rightarrow 2Z$

(B) $W \rightarrow X + Z$

(C) $W \rightarrow 2Y$

(D) $X \rightarrow Y + Z$

Q.3

Two deuterons undergo nuclear fusion to form a Helium nucleus. Energy released in this process is : (given binding energy per nucleon for deuteron $= 1.1\mathrm{MeV}$ and for helium $= 7.0\mathrm{MeV}$)

(A) $30.2\mathrm{MeV}$

(B) $32.4\mathrm{MeV}$

(C) $23.6\mathrm{MeV}$

(D) $25.8\mathrm{MeV}$

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Q.4

A reactor converts 100% of given mass into energy and that it operates at a power of 9×10^7 watt. The mass of the fuel consumed in 30 min in the reactor will be :

(A) $12 \times 10^{-3}\ \mathrm{kg}$

(B) $25 \times 10^{8}\ \mathrm{kg}$

(C) $18 \times 10^{-7}\ \mathrm{kg}$

(D) $11 \times 10^{-4}\ \mathrm{kg}$

Q.5

If a star can convert all the He nuclei completely into oxygen nuclei, the energy released per oxygen nuclei is [Mass of He nucleus is 4.0026 amu and mass of Oxygen nucleus is $15.9994\mathrm{amu}$]

(A) $7.6\mathrm{MeV}$

(B) $56.12\mathrm{MeV}$

(C) $10.24\mathrm{MeV}$

(D) $23.9\mathrm{MeV}$

Q.6

In the sun about 4 billion kg of matter is converted to energy each second. The power output of the sun in watt is

(A) 3.6×10^{26}

(B) 0.36×10^{26}

(C) 36×10^{26}

(D) 0.036×10^{26}

Q.7

Imagine that a reactor converts all given mass into energy and that it operates at a power level of 10^9 watt. The mass of the fuel consumed per hour in the reactor will be : (velocity of light, c is $3 \times 10^8\ \mathrm{m/s}$)

(A) $0.96\mathrm{gm}$

(B) $0.8\mathrm{gm}$

(C) $4 \times 10^{-2}\mathrm{gm}$

(D) $6.6 \times 10^{-5}\mathrm{gm}$

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Q.8

Consider the following reaction ${}_1\mathrm{H}^2 + {}_1\mathrm{H}^2 \rightarrow {}_2\mathrm{He}^4 + \mathrm{Q}$. If
$\mathrm{m}\left({}_1\mathrm{H}^2\right) = 2.014\mathrm{lamu}$
$\mathrm{m}\left({}_2\mathrm{He}^4\right) = 4.0024\mathrm{amu}$. The energy Q released
(in MeV) in this fusion reaction is

(A) 12

(B) 6

(C) 24

(D) 48

Q.9

If $200\mathrm{MeV}$ energy is released in the fission of a single U^{235} nucleus, the number of fissions required per second to produce 1 kilowatt power shall be $\left(\right.$ Given $1\mathrm{eV} = 1.6 \times 10^{-19}\ \mathrm{J}\left.\right)$

(A) 3.125×10^{13}

(B) 3.125×10^{14}

(C) 3.125×10^{15}

(D) 3.125×10^{16}

Q.10

A nucleus with mass number 220 initially at rest emits an α -particle. If the Q value of the reaction is $5.5\mathrm{MeV}$, then the kinetic energy of the α -particle will be

(A) $4.4\mathrm{MeV}$

(B) $5.4\mathrm{MeV}$

(C) $5.6\mathrm{MeV}$

(D) $6.5\mathrm{MeV}$

Q.11

An \alpha-particle of energy $5\mathrm{MeV}$ is scattered through 180° by a fined uranium nucleus. The distance of closest approach is of the order of

(A) 10^{-12} cm

(B) 10^{-10} cm

(C) $1\mathring{A}$

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(D) 10^{-15} cm

Q.12

When U^{238} nucleus originally at rest, decays by emitting an alpha particle having a speed n, the recoil speed of the residual nucleus is

(A) $\frac{8 \mathrm{u}}{238}$

(B) $-\frac{4u}{234}$

(C) $\frac{8 \mathrm{u}}{234}$

(D) $-\frac{4u}{238}$

Q.13

If M_0 is the mass of an oxygen isotope ${}_8\mathrm{O}^{17}$, M_P and M_N are the masses of a proton and a neutron respectively, the nuclear binding energy of the isotope is

(A) $\mathrm{M}_0 \mathrm{C}^2$

(B) $\left(\mathrm{Mo}-17 \mathrm{M}_\mathrm{N}\right) \mathrm{C}^2$

(C) $\left(\mathrm{M}_0-8 \mathrm{M}_\mathrm{P}\right) \mathrm{C}^2$

(D) none of these

Q.14

If the binding energy per nucleon in Li^7 and He^4 nuclei are respectively $5.60 \mathrm{MeV}$ and $7.06 \mathrm{MeV}$, then energy of reaction $\mathrm{Li}^7+\mathrm{p} \rightarrow 2{}_2\mathrm{He}^4$ is

(A) $19.6 \mathrm{MeV}$

(B) $2.4 \mathrm{MeV}$

(C) $8.4 \mathrm{MeV}$

(D) $17.3 \mathrm{MeV}$

Q.15

Two radioactive nuclei, P and Q, are present in a given sample and both of them decay into a stable nuclei R with different decay constants. At time $t=0$, the number of active nuclei of P are $4N_0$ and that of Q are N_0. Half-life of P is 1 minute, where as that of Q is 2 minutes. Initially, there are no nuclei of R present in the sample. When the number of nuclei of P and Q become equal, then the number of nuclei of R present in the sample would be

(A) $\frac{5N_0}{2}$

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(B) $2N_0$

(C) $3N_0$

(D) $\frac{9N_0}{2}$

Answers & Solutions

Q.1 Answer:

10^{12} s

Solution:

Let the mass of $^3_1\mathrm{H}$ be x amu. Then the mass defect in the first process is

$$(\Delta m)_1 = m\left(^2_1\mathrm{H}\right) + m\left(^2_1\mathrm{H}\right) - m\left(^3_1\mathrm{H}\right) - m(p)$$
$$= 2.014 + 2.014 - x - 1.007 = (3.021 - x)\mathrm{u}$$

Mass defect in the second process will be

$$(\Delta m)_2 = m\left(^2_1\mathrm{H}\right) + m\left(^3_1\mathrm{H}\right) - m\left(^4_2\mathrm{H}\right) - m(n)$$
$$= 2.014 + x - 4.001 - 1.008 = (x - 2.995)\mathrm{u}$$

$\therefore$ Total mass defect is

$$\Delta m = (\Delta m)_1 + (\Delta m)_2 = (3.021 - x) + (x - 2.995)$$
$$= 0.026\mathrm{u}$$

Now $1\mathrm{amu} = 931\mathrm{MeV} = 931 \times 10^6 \times 1.6 \times 10^{-19}$ J.

Hence, energy released is given by

$$E = 0.026 \times 931 \times 10^6 \times 1.6 \times 10^{-19}$$
$$= 3.87 \times 10^{-12}\ \mathrm{J}$$

Now power $P = 10^{16}$ W. **Therefore, the number of deutrons per second is** $\frac{10^{16}}{3.87 \times 10^{-12}} = 2.58 \times 10^{27}$

per second

$\therefore$ Deutron supply will exhaust in $\frac{10^{40}}{2.58 \times 10^{27}} = 3.87 \times 10^{12}$ **seconds**

Hence the correct choice is (c).

Q.2 Answer:

$W \rightarrow 2Y$

Solution:

The binding energies of the reactants and the products in the given nuclear reactions are as follows:

\begin{array}{lcc} \ \hline \text { Reaction } & \text { Reactant } & \text { Products } \\ \ \hline \text { (a) } Y \rightarrow 2 \mathrm{Z} & 60 \times 8.5=510 \mathrm{MeV} & 2 \times 30 \times 5.0=300 \mathrm{MeV} \\ \ \text { (b) } \mathrm{W} \rightarrow \mathrm{X}+\mathrm{Z} & 120 \times 7.5=900 \mathrm{MeV} & (90 \times 8.0=30 \times

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5.0)=870 \mathrm{MeV} \\ \ \text { (c) } \mathrm{W} \rightarrow 2 \mathrm{Y} & 120 \times 7.5=900 \mathrm{MeV} & 2 \times 60 \times 8.5=1020 \mathrm{MeV} \\ \ \text { (d) } \mathrm{X} \rightarrow \mathrm{Y}+\mathrm{Z} & 90 \times 8.0=720 \mathrm{MeV} & (60 \times 8.5+30 \times 5.0)=660 \mathrm{MeV} \\ \ \hline \ \end{array}

The binding energy of the products in reaction (c) is greater than that of the reactant. Hence reaction (c) releases energy.

Q.3 Answer:

$23.6\mathrm{MeV}$

Solution:

$_1\mathrm{H}^2 + {}_1\mathrm{H}^2 \rightarrow {}_2\mathrm{He}^4$

Total binding energy of two deuterium nuclei $= 1.1 \times 4 = 4.4\mathrm{MeV}$

Binding energy of a $\left({}_2\mathrm{He}^4\right)$ **nuclei** $= 4 \times 7 = 28\mathrm{MeV}$

Energy released in this process $= 28 - 4.4 = 23.6\mathrm{MeV}$

Q.4 Answer:

$18 \times 10^{-7}\ \mathrm{kg}$

Solution:

the reactor will conume $\frac{9 \times 10^7}{9 \times 10^{16}} \times 1800\ \mathrm{kg}$

$= 18 \times 10^{-7}\ \mathrm{kg}$

Q.5 Answer:

$10.24\mathrm{MeV}$

Solution:

$4\ {}_2^4\mathrm{He} \longrightarrow {}_8^{16}\mathrm{O}$

B.E. $= \Delta m \times 931.5\mathrm{MeV}$

$= (4 \times 4.0026 - 15.9994) \times 931.5 = 10.24\mathrm{MeV}$

Q.6 Answer:

0.36×10^{26}

Solution:

$\frac{\mathrm{m}}{\mathrm{t}} = 4 \times 10^8 \mathrm{kgs}^{-1}$

$\mathrm{E} = \mathrm{mc}^2$

$\Rightarrow \frac{\mathrm{E}}{\mathrm{t}} = \left(\frac{\mathrm{m}}{\mathrm{t}}\right) \mathrm{c}^2 \Rightarrow \frac{\mathrm{E}}{\mathrm{t}} = 4 \times 10^8 \times 9 \times 10^{16}$

$\Rightarrow \frac{\mathrm{E}}{\mathrm{t}} = 3.6 \times 10^{25} \mathrm{Js}^{-1} \Rightarrow \frac{\mathrm{E}}{\mathrm{t}} = 3.6 \times 10^{25}\ \mathrm{W}$

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Q.7 Answer:

$4 \times 10^{-2}\text{gm}$

Solution:

Power level of reactor, $\mathrm{P} = \frac{\mathrm{E}}{\Delta \mathrm{t}} = \frac{\Delta \mathrm{mc}^2}{\Delta \mathrm{t}}$

mass of the fuel consumed per hour in the reactor,

$\frac{\Delta \mathrm{m}}{\Delta \mathrm{t}} = \frac{\mathrm{P}}{\mathrm{c}^2} = \frac{10^9}{(3\times 10^8)^2}\ \mathrm{kg/sec} = \frac{3600\times 10^9}{9\times 10^{16}} \times 10^3\ \frac{\mathrm{gm}}{\mathrm{hr}}$

Q.8 Answer:

24

Solution:

$_1H^2 + {}_1H^2 \rightarrow {}_2He^4 + Q$

$\Rightarrow \Delta \mathrm{m} = \mathrm{m}\left({}_2\mathrm{He}^4\right) - 2\ \mathrm{m}\left({}_1\mathrm{H}^2\right)$

$\Rightarrow \Delta \mathrm{m} = 4.0024 - 2(2.0141)$

$\Rightarrow \Delta \mathrm{m} = -0.0258\text{amu}$

Since, $Q = c^2 \Delta m$

$\Rightarrow \mathrm{Q} = (0.0258)(931.5)\text{MeV} \Rightarrow \mathrm{Q} = 24\text{MeV}$

Q.9 Answer:

3.125×10^{13}

Solution:

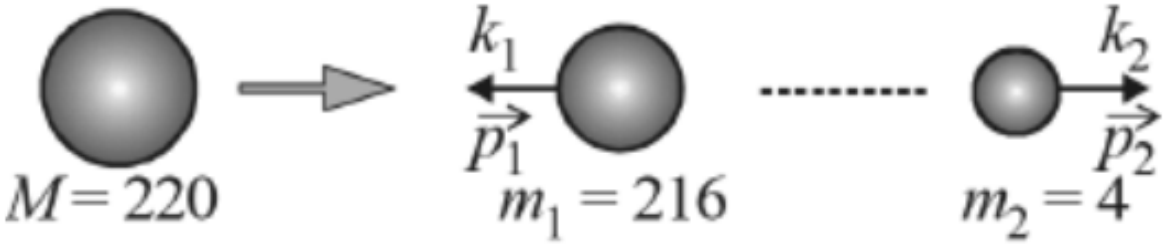
$$\mathrm{P} = \mathrm{n}\left(\frac{\mathrm{E}}{\mathrm{t}}\right) \Rightarrow 1000 = \frac{\mathrm{n} \times 200 \times 10^6 \times 1.6 \times 10^{-19}}{\mathrm{t}}$$

$$\Rightarrow \frac{\mathrm{n}}{\mathrm{t}} = 3.125 \times 10^{13}$$

Q.10 Answer:

5.4MeV

Solution:



Q-value of the reaction is 5.5MeV

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i.e., $k_1 + k_2 = 5.5MeV$**.....(i)**

By conservation of linear momentum

$p_1 = p_2 \Rightarrow \sqrt{2(216)k_1} = \sqrt{2(4)k_2}$

$\Rightarrow k_2 = 54k_1 \quad \ldots\ldots$**(ii)**

On solving equation (i) and (ii) we get

$k_2 = 5.4MeV$

Q.11 Answer:

10^{-12} cm

Solution:

At the closest apporach total energy is in the form of potential energy

$\therefore 5 \times 10^6 \times 1.6 \times 10^{-19} = \frac{Qq}{4\pi E_0 r}$

$= \frac{9\times10^9\times92\times1.6\times10^{-19}\times2\times1.6\times10^{-19}}{r}$

$= \frac{9\times10^9\times92\times2\times\left(1.6\times10^{-19}\right)^2}{5\times10^6\times16\times10^{-09}}$

$= \frac{9\times10^9\times92\times2\times1.6\times10^{-19}}{5\times10^6} \times 2.5 \times 10^{-12}$ cm

Q.12 Answer:

$-\frac{4u}{234}$

Solution:

According to principle of conservation of linear momentum the momentum of the system remains

the same before and after the decay.

Atomic mass of uraninm $= 238$ **and after emitting an alpha particle.**

$= 238 - 4 = 234$

$\therefore 238 \times 0 = 4u + 234v$

$\therefore v = -\frac{4u}{234}$

Q.13 Answer:

none of these

Solution:

B.E. $= E = \Delta mc^2$

$\Delta m =$ **mass defect**

$\Delta m = 8\ m_p + 9m_n - M_o$

B.E. $= (8m_p + 9m_n - M_O)\, c^2$

Q.14 Answer:

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17.3MeV

Solution:

B.E. of $\text{L}^7 = 39.20\text{MeV}$ **and**

$\text{He}^4 = 28.24\text{MeV}$

Hence binding energy of

$2\text{He}^4 = 56.48\text{MeV}$

Energy of reaction

$= 56.48 - 39.20 = 1728\text{McV}$

Q.15 Answer:

$\frac{9N_0}{2}$

Solution:

Initially, $P \to 4N_0; \quad Q \to N_0$

Half life $T_P = 1 \quad \text{min}; \quad T_Q = 2 \quad \text{min}$

Let after time t**, number of nuclei of** P **and** Q **are equal.**

That is $\frac{4N_0}{2^{t/1}} = \frac{N_0}{2^{t/2}}$

or $\frac{4}{2^{t/2}} = I$

or $t = 4 \quad \text{min}$

At $t = 4 \quad \text{min}$**,**

$N_P = \frac{4N_0}{2^{4/1}} = \frac{N_0}{4}$

At $t = 4 \quad \text{min}, \quad N_Q = \frac{N_0}{2^{4/2}} = \frac{N_0}{4}$ **or number of nuclei of** $R = 4N_0 - \frac{N_0}{4} + N_0 - \frac{N_0}{4} = \frac{9N_0}{2}$

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